

# Repository GitHub — Workshop Flutter Avancé

## Structure complète du projet

```
flutter-advanced-workshop/
|
├─ README.md
├─ .gitignore
├─ analysis_options.yaml
├─ pubspec.yaml
├─ firebase.json
├─ firestore.rules
├─ .env.example
|
├─ assets/
│   ├── images/
│   │   ├── logo.png
│   │   ├── empty.png
│   │   └── chatbot.png
│   └── animations/
│       └── loading.json
|
├─ lib/
│   ├── main.dart
│   ├── core/
│   │   ├── constants/
│   │   │   ├── app_colors.dart
│   │   │   ├── app_theme.dart
│   │   │   └── api_constants.dart
│   │   ├── utils/
│   │   │   ├── validators.dart
│   │   │   ├── extensions.dart
│   │   │   └── logger.dart
│   │   └── services/
│   │       ├── auth_service.dart
│   │       ├── firestore_service.dart
│   │       ├── api_service.dart
│   │       ├── cache_service.dart
│   │       └── ai_service.dart
│   └── data/
│       ├── models/
│       │   └── user_model.dart
```

```

├── task_model.dart
├── post_model.dart
├── message_model.dart
├── repositories/
│   ├── auth_repository.dart
│   ├── task_repository.dart
│   ├── post_repository.dart
│   └── chat_repository.dart
├── providers/
│   ├── auth_provider.dart
│   ├── task_provider.dart
│   ├── post_provider.dart
│   └── theme_provider.dart
├── presentation/
│   ├── pages/
│   │   ├── splash_page.dart
│   │   ├── login_page.dart
│   │   ├── signup_page.dart
│   │   ├── home_page.dart
│   │   ├── posts_page.dart
│   │   ├── chatbot_page.dart
│   │   └── settings_page.dart
│   ├── widgets/
│   │   ├── custom_textfield.dart
│   │   ├── task_card.dart
│   │   ├── loading_widget.dart
│   │   ├── post_tile.dart
│   │   ├── empty_widget.dart
│   │   └── message_bubble.dart
│   └── navigation/
│       ├── app_router.dart
│       └── routes.dart
├── firebase_options.dart
├── test/
│   ├── unit/
│   │   ├── post_model_test.dart
│   │   ├── auth_repository_test.dart
│   │   └── cache_service_test.dart
│   └── widget/
│       ├── login_page_test.dart
│       ├── home_page_test.dart
│       └── chatbot_page_test.dart

```

```
|
|
|─ docs/
|   │─ workshop_slides.pdf
|   │─ architecture.png
|   └─ setup_guide.md
|
|─ screenshots/
|   │─ login.png
|   │─ home.png
|   │─ posts.png
|   └─ chatbot.png
```

---

## README.md

```
# Flutter Advanced Workshop

Workshop interactif Flutter :
- Firebase Auth
- Firestore CRUD
- API REST
- Cache local Hive
- Performance Flutter
- Material 3
- IA Chatbot
- Tests unitaires et widgets

## Technologies

- Flutter 3.x
- Firebase
- Firestore
- Provider
- Hive
- HTTP
- Material 3

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# Installation

## 1. Cloner le projet

```bash

git clone https://github.com/username/flutter-advanced-workshop.git
cd flutter-advanced-workshop
```

## 2. Installer les dépendances

```
flutter pub get
```

## 3. Configurer Firebase

```
dart pub global activate flutterfire_cli  
flutterfire configure
```

## 4. Lancer le projet

```
flutter run
```

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# Fonctionnalités

## Authentification Firebase

- Signup
- Login
- Logout
- Session persistante

## Firestore

- CRUD temps réel
- Synchronisation automatique

## API REST

- Consommation API
- Gestion erreurs
- FutureBuilder

## Cache Offline

- Hive
- Offline First

## UI

- Material 3
- Dark mode
- Navigation moderne

## IA

- Mini chatbot

## Tests

- Tests unitaires
  - Tests widgets
- 

## Structure Architecture

```
Presentation
  ↓
Provider
  ↓
Repository
  ↓
Services
  ↓
Firebase / API
```

---

## Captures

Login   Home

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## Workshop

Projet réalisé dans le cadre du cours : Développement Multiplateforme — Flutter

```
---

# pubspec.yaml

``yaml
name: flutter_advanced_workshop

description: Workshop Flutter avancé

publish_to: 'none'

version: 1.0.0+1
```

```
environment:
  sdk: '>=3.0.0 <4.0.0'

dependencies:
  flutter:
    sdk: flutter

  cupertino_icons: ^1.0.6

  firebase_core: ^2.24.2
  firebase_auth: ^4.15.3
  cloud_firestore: ^4.13.6

  provider: ^6.1.1

  http: ^1.1.2

  hive: ^2.2.3
  hive_flutter: ^1.1.0

  flutter_dotenv: ^5.1.0

  lottie: ^2.7.0

  google_fonts: ^6.1.0

  flutter_spinkit: ^5.2.0

  intl: ^0.18.1

dev_dependencies:
  flutter_test:
    sdk: flutter

  flutter_lints: ^3.0.1

flutter:
  uses-material-design: true

  assets:
    - assets/images/
    - assets/animations/
    - .env
```

## .gitignore

```
.dart_tool/  
.packages  
.pub/  
build/  
android/.gradle/  
ios/Pods/  
  
.env  
firebase_options.dart  
  
.idea/  
.vscode/
```

## firebase.rules

```
rules_version = '2';  
service cloud.firestore {  
  match /databases/{database}/documents {  
  
    match /tasks/{taskId} {  
      allow read, write: if request.auth != null;  
    }  
  }  
}
```

## main.dart

```
import 'package:flutter/material.dart';  
import 'package:firebase_core/firebase_core.dart';  
import 'firebase_options.dart';  
  
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  
  await Firebase.initializeApp(  
    options: DefaultFirebaseOptions.currentPlatform,  
  );  
  
  runApp(const MyApp());  
}
```

```

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Flutter Workshop',
      theme: ThemeData(
        useMaterial3: true,
        colorSchemeSeed: Colors.blue,
      ),
      home: const Scaffold(
        body: Center(
          child: Text('Flutter Workshop'),
        ),
      ),
    );
  }
}

```

## Branches Git recommandées

```

main
|
├─ step-0-setup
├─ step-1-auth
├─ step-2-firestore
├─ step-3-api
├─ step-4-cache
├─ step-5-ui
├─ step-6-ai
└─ final-project

```

## Workflow pédagogique recommandé

### Avant le workshop

Le formateur prépare :

- Projet Firebase
- API key
- Repo GitHub



- Branches step-by-step

## Pendant le workshop

Les étudiants : 1. clonage repo 2. checkout branche 3. réalisation défis 4. commit progression

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## Convention Git

```
git checkout -b feature/login  
  
git add .  
  
git commit -m "Add Firebase login"  
  
git push origin feature/login
```

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## Releases GitHub recommandées

```
v1.0 - Setup Flutter  
v1.1 - Firebase Auth  
v1.2 - Firestore CRUD  
v1.3 - API REST  
v1.4 - Offline Cache  
v1.5 - Material 3  
v1.6 - AI Chatbot  
v2.0 - Final Workshop
```

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## GitHub Actions (CI Flutter)

Créer : .github/workflows/flutter.yml

```
name: Flutter CI  
  
on:  
  push:  
    branches: [ main ]  
  
jobs:  
  build:  
    runs-on: ubuntu-latest
```

```
steps:
  - uses: actions/checkout@v3

  - uses: subosito/flutter-action@v2
    with:
      flutter-version: '3.16.0'

  - run: flutter pub get

  - run: flutter analyze

  - run: flutter test
```

---

## Arborescence pédagogique recommandée

### Pour les étudiants

```
step-0 = setup
step-1 = auth
step-2 = firestore
step-3 = api
step-4 = cache
step-5 = ui
step-6 = ai
final = solution complète
```

---

## Bonus : organisation des commits

```
feat(auth): add firebase login
feat(api): add REST service
feat(cache): implement Hive storage
feat(ai): add chatbot page
fix(ui): improve loading state
refactor(home): split widgets
```

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## Déploiement futur possible

- Firebase Hosting
- Android APK
- Play Store
- Web Flutter

## Ressources utiles

### Flutter

<https://flutter.dev>

### Firebase

<https://firebase.google.com>

### Provider

<https://pub.dev/packages/provider>

### Hive

<https://pub.dev/packages/hive>

### HTTP

<https://pub.dev/packages/http>

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## Résultat final attendu

Application Flutter complète avec :

✓ Firebase Auth ✓ Firestore temps réel ✓ API REST ✓ Cache Offline ✓ Material 3 ✓ Dark mode ✓  
Architecture propre ✓ Provider ✓ Tests ✓ Chatbot IA ✓ Déploiement GitHub